

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

One-variable data: Distributions and measures of center and spread

01

10 questions · ~15 min

Q1

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

International Tourist
Arrivals, in millions

Country **2012** **2013**

France 83.0 84.7

**United
States** 66.7 69.8

Spain 57.5 60.7

China 57.7 55.7

Italy 46.4 47.7

Turkey 35.7 37.8

Germany 30.4 31.5

**United
Kingdom** 26.3 32.2

Russia 24.7 28.4

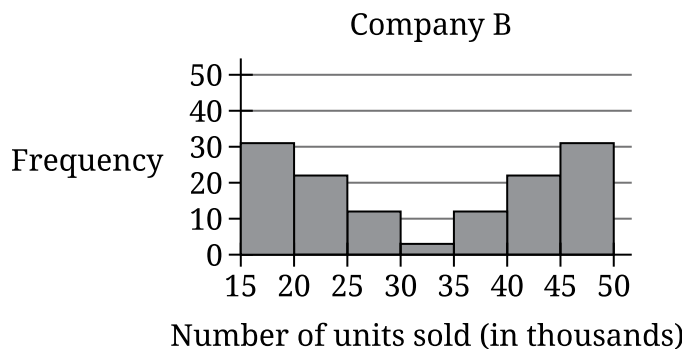
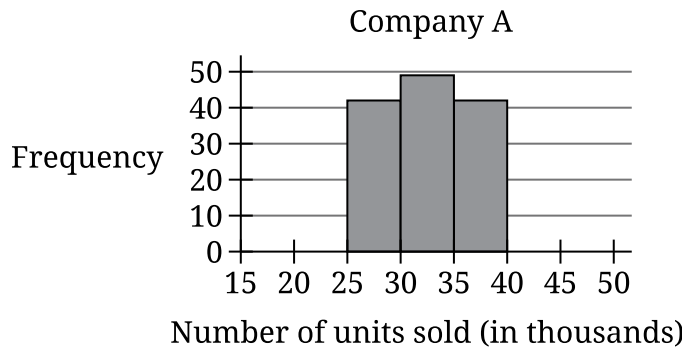
The table above shows the number of international tourist arrivals, rounded to the nearest tenth of a million, to the top nine tourist destinations in both 2012 and 2013. Based on the information given in the table, how much greater, in millions, was the median number of international tourist arrivals to the top nine tourist destinations in 2013 than the median number in 2012, to the nearest tenth of a million?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5

6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Company A:
 - The horizontal axis is labeled Number of units sold (in thousands). It ranges from 15 to 50 and is divided into 7 equal intervals.
 - The vertical axis is labeled Frequency. It ranges from 0 to 50.
 - The histogram has 3 bins.
 - The Frequency data for the 3 bins are as follows:
 - 25 to 29: 43
 - 30 to 34: 48
 - 35 to 39: 43
- Company B:
 - The horizontal axis is labeled Number of units sold (in thousands). It ranges from 15 to 50 and is divided into 7 equal intervals.
 - The vertical axis is labeled Frequency. It ranges from 0 to 50.
 - The histogram has 7 bins.
 - The Frequency data for the 7 bins are as follows:

- 15 to 19: 31
- 20 to 24: 24
- 25 to 29: 11
- 30 to 34: 3
- 35 to 39: 11
- 40 to 44: 24
- 45 to 49: 31

The histograms summarize the distributions of number of units sold, in thousands, for company A and company B. Which statement best compares the standard deviations of number of units sold for these companies?

- A. The standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.
- B. The standard deviation of number of units sold for company A is greater than the standard deviation of number of units sold for company B.
- C. The standard deviation of number of units sold for company A is equal to the standard deviation of number of units sold for company B.
- D. There is not enough information to compare the standard deviations.

The results of two independent surveys are shown in the table below.

Men's Height

Group	Sample size	Mean (centimeters)	Standard deviation (centimeters)
A	2,500	186	12.5
B	2,500	186	19.1

Which statement is true based on the table?

- A. The Group A data set was identical to the Group B data set.
- B. Group B contained the tallest participant.
- C. The heights of the men in Group B had a larger spread than the heights of the men in Group A.
- D. The median height of Group B is larger than the median height of Group A.

Ages of 20 Students Enrolled in a College Class

Age	Frequency
18	6
19	5
20	4
21	2
22	1
23	1
30	1

The table above shows the distribution of ages of the 20 students enrolled in a college class. Which of the following gives the correct order of the mean, median, and mode of the ages?

- A. mode < median < mean
- B. mode < mean < median
- C. median < mode < mean
- D. mean < mode < median

Q5

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The table shows the frequency of values in a data set.

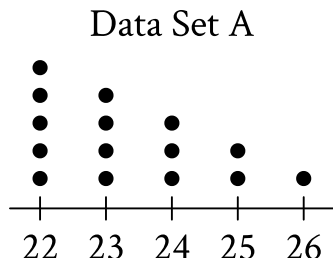
Value	Frequency
19	7
21	1
23	7
25	4

What is the minimum value of the data set?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The data for the dot plot are as follows:
 - 22: 5 dots
 - 23: 4 dots
 - 24: 3 dots
 - 25: 2 dots
 - 26: 1 dot

Data set A has 15 values and is represented by the dot plot shown. Data set B is created by adding 46 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

- A. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
- B. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.
- C. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
- D. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

The weights, in pounds, for 15 horses in a stable were reported, and the mean, median, range, and standard deviation for the data were found. The horse with the lowest reported weight was found to actually weigh 10 pounds less than its reported weight. What value remains unchanged if the four values are reported using the corrected weight?

- A. Mean
- B. Median
- C. Range
- D. Standard deviation

Station 1	Station 2	Station 3	Station 4	Station 5
\$3.699	\$3.609	\$3.729	\$3.679	\$3.729

In the table above, Melissa recorded the price of one gallon of regular gas from five different local gas stations on the same day. What is the median of the gas prices Melissa recorded?

- A. \$3.679
- B. \$3.689
- C. \$3.699
- D. \$3.729

Q9

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2, 10, 3, 7, 6

The mean of the list of numbers above is what fraction of the sum of the five numbers?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

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If a is the mean and b is the median of nine consecutive integers, what is the value of $|a - b|$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.